Polynomials — problem 10.8

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The polynomial $f(x, y, z)$ is antisymmetric if the sign changes on switching any two variables. Prove that every antisymmetric polynomial $f(x, y, z)$ can be written in the form $f(x, y, z) = (x - y)(x - z)(y - z)g(x, y, z)$, where $g(x, y, z)$ is symmetric.